

Aviation Wildlife Strikes: An Analysis of Wildlife Strikes Across Airports Throughout the United States

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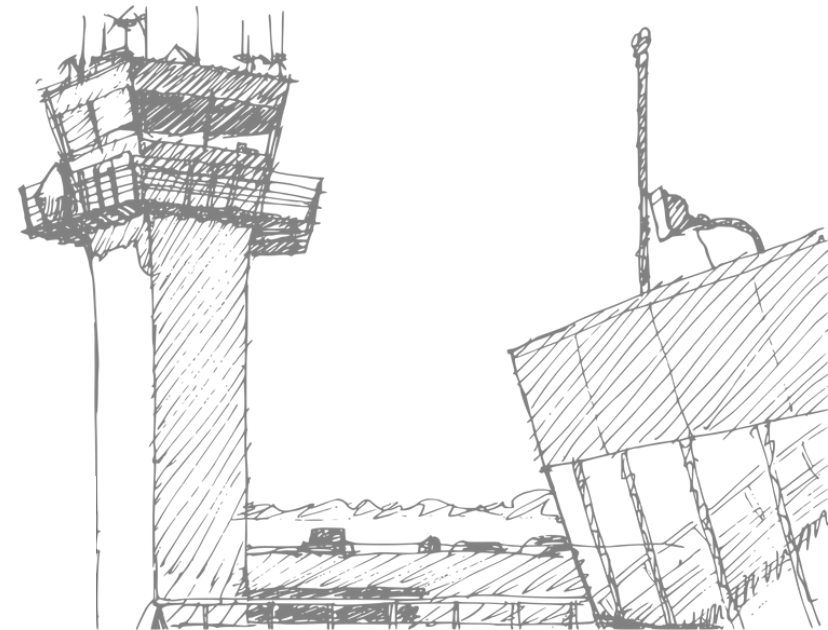


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Purpose & Beneficiaries

Wildlife strikes – collisions between aircraft and birds or other animals – are one of aviation's most persistent and underappreciated safety challenges, biological in nature, unpredictable in timing, and directly tied to geography and migration patterns. Since 1990, the Federal Aviation Administration has maintained the National Wildlife Strike Database, a voluntary reporting system that has grown into one of the most comprehensive aviation safety records in the world, capturing 342,890 reported incidents across 713 US airports over 35 years.

This report presents an independent analysis of the full FAA Wildlife Strike Database, covering January 1990 through April 2026.

This is designed to surface three actionable insights: which airports face the highest strike frequency, during which flight phase strikes are most dangerous, and how strike volume and costs have evolved over 35 years.

The analysis was conducted using SQL-based data exploration in JetBrains DataGrip and visualized through an interactive Microsoft Power BI dashboard. All findings are derived directly from FAA-reported data with no modifications to the underlying records.

This analysis is relevant to a broad audience across the aviation safety ecosystem:

Airport Operations & Safety Teams – Strike concentration data by airport and geographic location directly informs where wildlife hazard management resources are most needed, and which flight corridors warrant the greatest attention.

Aviation Safety Professionals & Regulators – The 35-year trend analysis provides longitudinal context for evaluating the effectiveness of existing wildlife mitigation programs and identifying periods where reporting behavior or actual strike frequency shifted.

Airlines & Fleet Operators – Phase-of-flight risk data helps frame operational risk during approach, take-off, and climb – the three phases that together account for nearly 75% of all recorded strikes.

Researchers & Data Analysts – The methodology and SQL queries used in this analysis are fully documented and publicly available, providing a reproducible foundation for further investigation into species-specific risk, seasonal patterns, or regional trends.

General Aviation Community – Understanding the scale, geography, and trend of wildlife strikes builds broader awareness of a hazard that affects every segment of flight operations, from commercial carriers to general aviation pilots operating at non-towered airports.

Executive Summary

Wildlife strikes represent one of the most consistently underreported cost centers in commercial aviation. Between 1990 and 2025, the Federal Aviation Administration recorded 342,890 strike incidents across 713 US airports – a figure that, by the FAA's own estimates, represents only a fraction of actual strike occurrences given the voluntary nature of reporting. Annual strike volume has tripled over 35 years, inflation-adjusted repair costs have exceeded \$95M in peak years, and the geographic and operational concentration of risk is both measurable and actionable.

This report presents three findings drawn directly from the full FAA National Wildlife Strike Database, each with direct implications for both private aviation operators and government safety agencies.

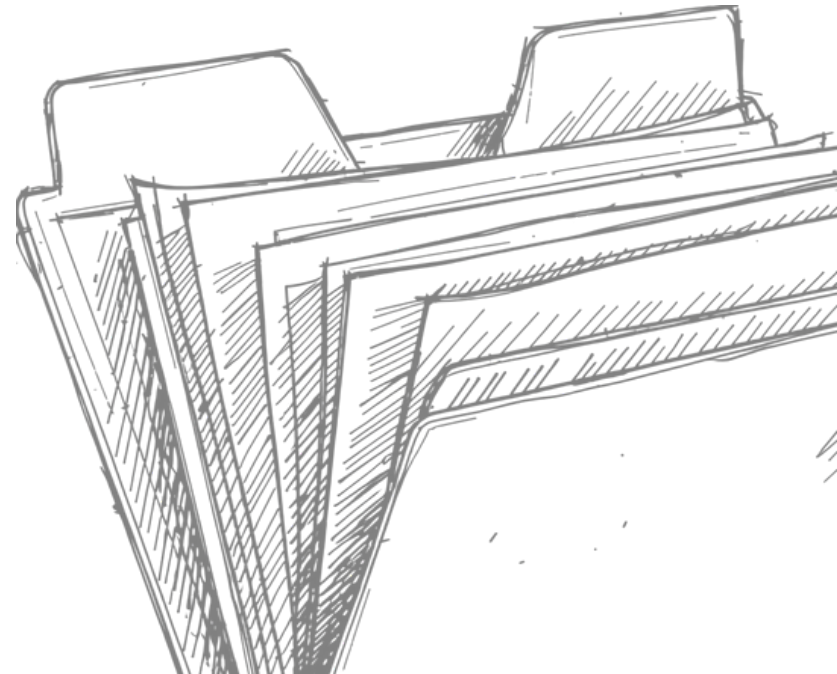
Data Overview

Statistic	Value
342,890	Total Strike Reports Analyzed
35 Years	Data Coverage (1990 – 2025)
713	US Airports Represented
23,000+	Strikes Recorded in 2024 Alone
~\$95M	Peak Single-Year Inflation-Adjusted Repair Cost (2009)

Section 2

The Findings

*Three Data-Driven Insights from
342,890 FAA Wildlife Strike Records*



Finding 1

Measurable & Addressable Airport Concentration



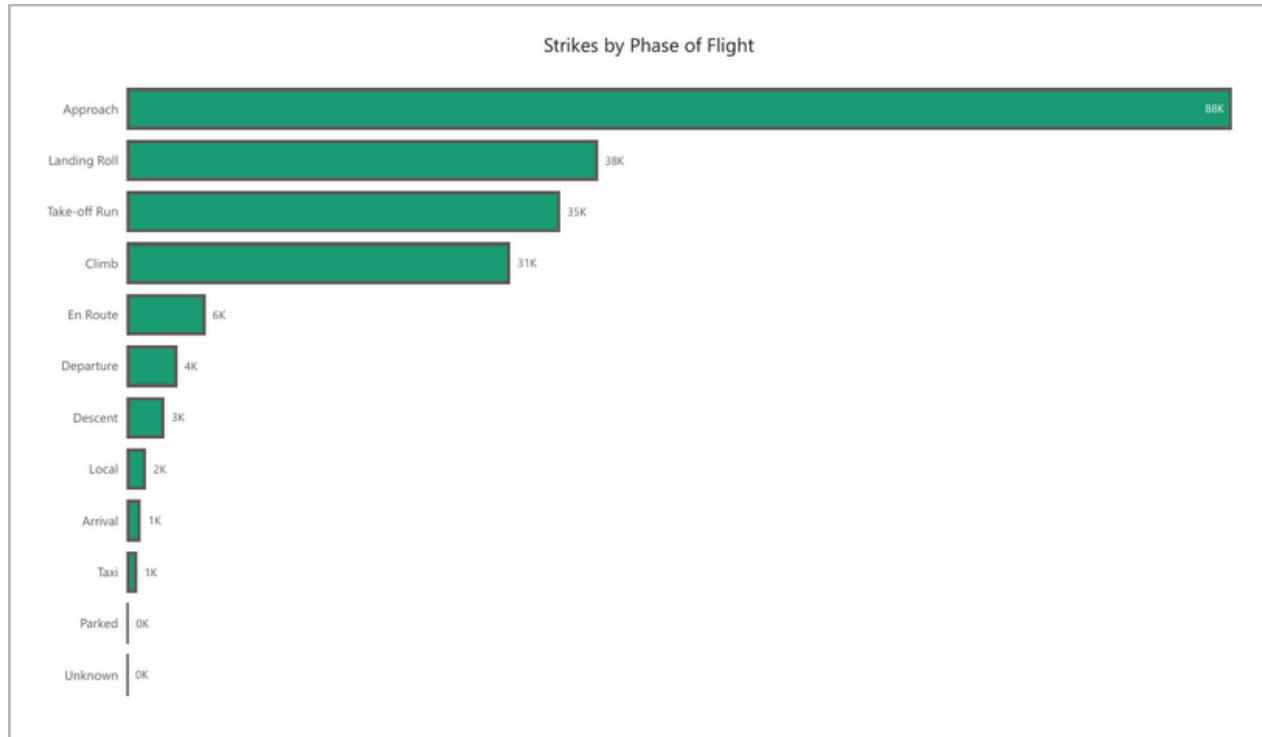
Denver International leads all US airports with 11,538 recorded strikes – nearly 30% more than Dallas/Fort Worth (8,889). The top 5 airports collectively represent a disproportionate share of total incidents, concentrated along major migratory bird corridors.

For Airport Operators & Airlines: High-frequency airports represent concentrated exposure to repair costs, schedule disruption, and AOG events. Strike frequency data provides a direct input for insurance risk modeling and wildlife hazard management investment prioritization.

For Regulators & Safety Agencies: Geographic concentration confirms that targeted habitat management and deterrence program investment at a small number of high-risk airports can produce outsized system-wide safety returns.

Finding 2

75% of Strikes Occur in Three Predictable Flight Phases



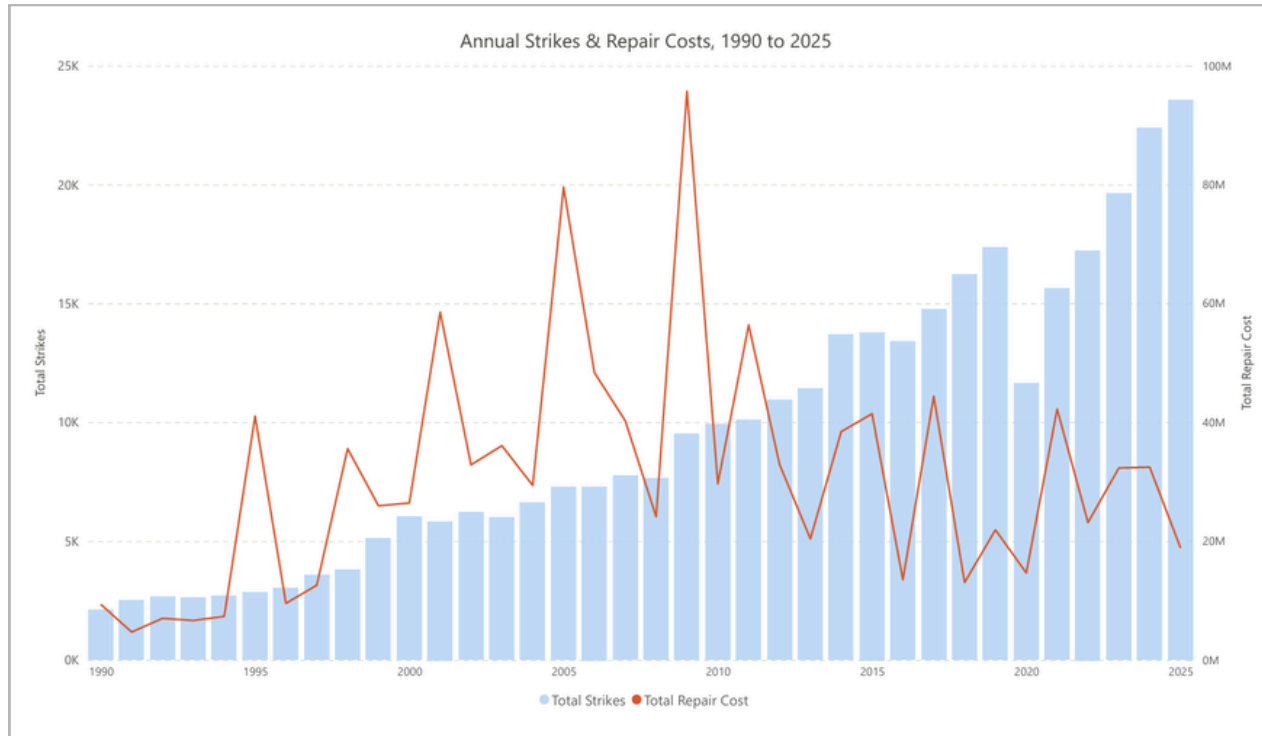
Approach alone accounts for 88,268 strikes – 25.7% of all recorded incidents. Together, approach, landing roll, and take-off run represent nearly 75% of the full dataset. Climb phase, while lower in volume, produces the highest rate of engine damage per strike.

For Airport Operators & Airlines: Phase concentration means risk is not random – it is bounded to specific operational windows that can be actively managed through crew awareness, approach corridor monitoring, and ground deterrence activation protocols.

For Regulators & Safety Agencies: The approach-phase dominance provides a clear mandate for wildlife hazard management programs to prioritize the 3-mile final approach corridor and immediate airport perimeter over broader geographic coverage.

Finding 3

Growing Strike Volume; Volatile & Concentrated Cost Impact



Annual strike volume has grown from ~2,000 in 1990 to over 23,000 in 2024. Repair costs are highly volatile – driven by a small number of high-severity engine ingestion events that generate outsized economic consequences relative to their frequency. The 2009 US Airways Flight 1549 incident remains the clearest single inflection point in both reporting behavior and industry awareness.

For Airport Operators & Airlines: Cost volatility is a tail-risk problem. A single high-severity strike season can generate repair and revenue loss costs that dwarf years of wildlife mitigation program investment – making the ROI case for proactive hazard management straightforward.

For Regulators & Safety Agencies: The post-2009 reporting surge demonstrates that high-profile incidents drive voluntary compliance. Sustained outreach, simplified reporting tools, and mandatory reporting expansion could significantly improve the dataset quality available for future risk modeling.

Conclusion

Wildlife strike risk in the United States is not a random or uncontrollable phenomenon – it is concentrated, measurable, and growing in ways that the data makes clearly visible. Over 35 years, the FAA's National Wildlife Strike Database has captured the steady rise of a hazard that now produces more than 23,000 reported incidents annually, with repair costs that can exceed \$95M in a single year and safety consequences that, in the most severe cases, put hundreds of lives at risk.

The three findings in this report point to the same underlying truth: that wildlife strike risk follows predictable patterns across geography, flight operations, and time. Denver International and a small cluster of high-frequency airports account for a disproportionate share of total incidents. Approach, landing roll, and take-off run together represent nearly 75% of all strikes – bounded to specific operational windows that are amenable to active management. And the long-term upward trend in both volume and cost shows no sign of reversing, driven by expanding air traffic, recovering wildlife populations, and airspace that grows busier every year.

For private operators – airlines, airport authorities, and maintenance organizations – the data makes a straightforward economic case. The cost of proactive wildlife hazard management is modest relative to the tail-risk exposure of a single high-severity engine ingestion event. Strike frequency data by airport and phase of flight provides a direct, evidence-based input for risk modeling, insurance planning, and operational protocol development.

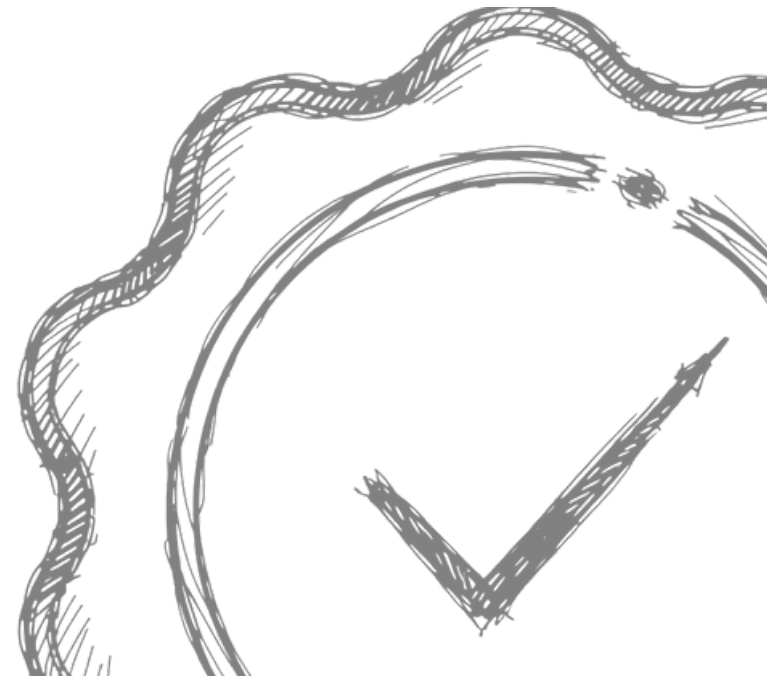
For government agencies and regulators – the FAA, USDA Wildlife Services, and their international counterparts – the data provides a longitudinal foundation for evaluating the effectiveness of existing mitigation programs and identifying where investment has the greatest potential to reduce system-wide risk. The post-2009 surge in reporting demonstrates that industry behavior responds to awareness. Sustained outreach, mandatory reporting expansion, and simplified submission tools could substantially improve the quality of the risk intelligence available to the entire aviation safety community.

The evidence is clear. The geography is known. The flight phases are identified. What the data ultimately calls for – from both the private and public sectors – is coordinated, targeted, and sustained action at the airports, corridors, and operational windows where the risk is highest and the potential for impact is greatest.

Section 3

Implications & Recommendations

*Turning the Three Findings Into
Targeted Action*



For Airport Operators & Airlines

- **Prioritize High-Frequency Airports for Wildlife Hazard Investment** – Denver International, Dallas/Fort Worth, Chicago O'Hare, JFK, and Memphis collectively represent the highest strike exposure in the US. Airports in this tier should conduct annual wildlife hazard assessments, maintain active deterrence programs year-round, and allocate dedicated budget for habitat modification within the airport perimeter and approach corridor. Airlines with heavy operations at these airports should factor strike frequency into route risk models and maintenance scheduling.
- **Concentrate Ground Deterrence on the Approach Corridor** – With 25.7% of all strikes occurring during approach, the 3-mile final approach path represents the single highest-priority zone for active wildlife deterrence. Ground-based radar systems, trained wildlife biologists, and habitat management targeting food sources and water features within this corridor offer the highest return on mitigation investment.
- **Model Tail-Risk Scenarios for Engine Ingestion Events** – Repair cost data shows that annual costs are driven by a small number of catastrophic events rather than the cumulative impact of minor strikes. Airlines and insurers should model worst-case scenarios where a single season produces multiple engine ingestion events at high-frequency airports – and ensure that maintenance reserves, AOG protocols, and insurance coverage reflect this tail-risk exposure.
- **Invest in Climb-Phase Monitoring** – While approach dominates in volume, climb phase produces the highest rate of engine damage per strike. Airlines operating at airports adjacent to large bird populations – particularly waterfowl and raptors – should consider enhanced climb corridor monitoring and crew awareness protocols for the first 3,000 feet of climb.

For Regulators & Government Safety Agencies

- **Expand Mandatory Reporting Requirements** – The voluntary nature of strike reporting means the database captures an estimated fraction of actual incidents. Expanding mandatory reporting – particularly for commercial carriers and Part 135 operators – would substantially improve the risk intelligence available to safety agencies, researchers, and airport operators. The post-2009 reporting surge demonstrates that the industry responds to clear expectations and simplified submission processes.
- **Concentrate Habitat Management Resources Geographically** – The top 5 airports account for a disproportionate share of total strikes. Directing USDA Wildlife Services resources, habitat modification funding, and biological expertise to these high-frequency locations – rather than distributing resources evenly across all airports – would maximize the system-wide safety return on government investment.
- **Evaluate Mitigation Program Effectiveness Using Longitudinal Data** – The 35-year dataset provides a unique foundation for evaluating whether specific interventions – habitat modification programs, deterrence technology deployments, or reporting campaigns – produced measurable reductions in strike frequency at targeted airports. Agencies should establish baseline metrics and conduct periodic longitudinal reviews to assess program effectiveness against the trend.
- **Strengthen International Data Sharing** – The FAA database captures strikes at 92 airports in 55 countries reported by US carriers – but international strike data remains fragmented across national aviation authorities. Strengthening data sharing agreements between the FAA, ICAO, and national civil aviation authorities would produce a more complete global picture of wildlife strike risk and enable coordinated mitigation strategies along shared migratory flyways.

Shared Priority – Species Identification & DNA Analysis

This analysis addresses strike frequency, phase-of-flight risk, and cost trends at the aggregate level – but the FAA Wildlife Strike Database contains significantly more analytical depth than what a single project can fully surface. The following areas represent natural extensions of this work for researchers, safety agencies, and operators seeking more granular risk intelligence.

Species-Level Risk Analysis – The database includes species identification fields for a substantial portion of records. A dedicated species-level analysis would identify which specific wildlife populations drive the highest damage rates at individual airports – moving beyond frequency counts to damage-weighted risk scores by species and location.

This work would pair naturally with data from the Smithsonian Institution's Feather Identification Laboratory, which conducts DNA-based species identification from strike remains for military and FAA-reported incidents.

Expanding forensic species identification to commercial aviation would produce site-specific biological intelligence that airport wildlife programs currently lack – enabling deterrence strategies targeting the right species in the right environments rather than relying on generalized regional data.

Seasonal and Migration Pattern Analysis – Strike frequency almost certainly follows seasonal patterns tied to bird migration cycles, breeding seasons, and winter roosting behavior. A month-by-month analysis overlaid with known migratory flyway data would allow airport operators to implement time-specific deterrence escalation protocols during peak risk windows rather than maintaining uniform year-round programs.

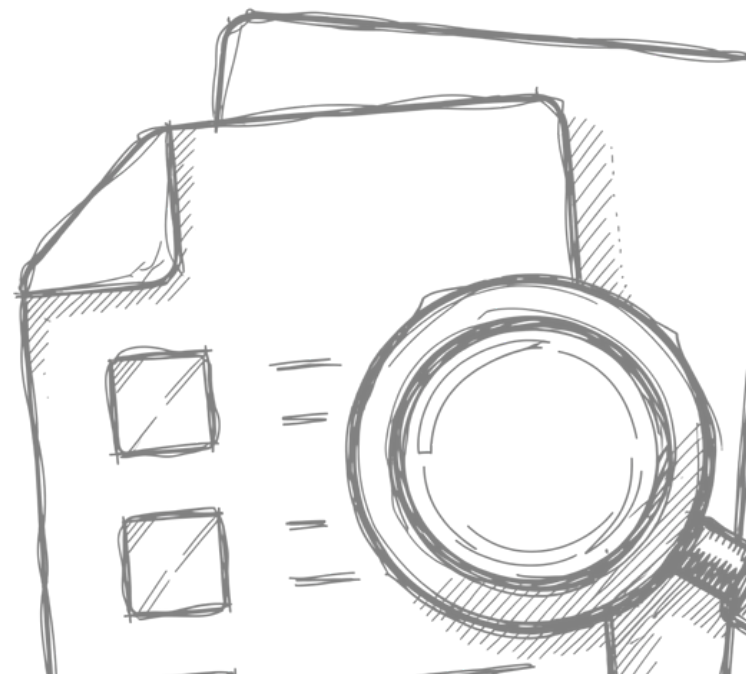
Damage Severity Modeling by Aircraft Type – The database includes aircraft type fields that would allow analysis of which airframes are most vulnerable to damage at equivalent strike energies – useful for fleet operators assessing relative exposure across mixed fleets and for manufacturers evaluating design modifications that limit damage from bird ingestion events.

International Strike Comparison – The FAA database captures strikes reported by US carriers at 92 airports across 55 countries. A comparative analysis of international strike rates against domestic figures would provide context for evaluating whether US wildlife hazard management programs are producing better outcomes than equivalent programs in other jurisdictions – and identify international best practices worth adopting domestically.

Section 4

Methodology

*How the Data Was Collected,
Processed, and Analyzed*



This analysis was conducted using a structured, end-to-end analytics workflow spanning data acquisition, verification, SQL-based exploration, and interactive visualization. All steps are fully documented and reproducible using publicly available tools and data.

Data Acquisition	The full FAA National Wildlife Strike Database was downloaded from wildlife.faa.gov as a Microsoft Excel file (.xlsx) containing 342,890 records across 102 columns, spanning January 1990 through April 2026. No filters were applied during download – the complete database was used as the source for all analysis.
Data Verification	Given the file size of 145MB, two independent CSV conversions were conducted – one via CloudConvert and one using Anthropic Claude – and cross-referenced against each other to confirm data completeness. The FAA's own uploaded CSV (342,890 rows) was confirmed as the authoritative source file. This verification step identified a 2,523-record discrepancy in one conversion caused by a known limitation in Python's openpyxl library with large Excel files – underscoring the importance of cross-referencing conversion outputs before analysis.
Data Storage & Querying	The verified CSV was imported into a SQLite database (wildlife.db) using JetBrains DataGrip. The full 342,890-row dataset was loaded into a single table named strikes under the main schema. All exploration and analysis was conducted using SQL.

Two rounds of queries were executed. The first round comprised data integrity checks – confirming row counts, checking for null values across key columns, reviewing year range coverage, and listing all distinct values in categorical fields. The second round comprised three targeted analysis queries producing the aggregated datasets used in the Power BI dashboard.

Exported Datasets Each analysis query was exported from DataGrip as a clean, pre-aggregated CSV file:

File	Contents	Rows
airports.csv	Strike counts by airport with latitude, longitude, state, and damaging strike count	30
phases.csv	Strike counts by phase of flight with minor, substantial, and destroyed damage breakdown	12
yearly_trend.csv	Annual strike volume and inflation-adjusted repair costs, 1990–2025	36

Exporting pre-aggregated results rather than the full source file kept the Power BI dashboard lightweight and ensured fast, responsive visual rendering.

Visualization	The three exported CSV files were loaded into Microsoft Power BI Desktop. Three visuals were built – a bubble map, a clustered horizontal bar chart, and a dual-axis line and column combo chart. All field mappings, filters, color schemes, axis formatting, and titles were configured manually. The dashboard was exported as a static PDF for distribution and published to Power BI Service for interactive access.
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Limitations

- Strike reporting is voluntary. The database represents reported incidents only – actual strike frequency is estimated to be higher than recorded figures across all years.
- Reporting rates increased significantly after the 2009 US Airways Flight 1549 incident. The post-2009 upward trend in strike volume partially reflects improved reporting behavior rather than a proportional increase in actual incidents.
- Repair cost data is incomplete for a substantial portion of records. Zero values in the cost field may represent unreported costs rather than no damage – meaning the repair cost figures in this analysis are likely conservative underestimates.
- The UNKNOWN airport category (45,719 strikes) was excluded from geographic analysis due to missing latitude and longitude data. These records are included in all non-geographic analyses.
- Species identification varies in accuracy across records. Older records in particular rely on visual identification rather than forensic analysis, introducing potential misclassification at the species level.

Tools & Technologies

Tool	Purpose
JetBrains DataGrip	SQL Querying, Data Exploration, & CSV Export
SQLite	Local Database Engine
Microsoft Power BI Desktop	Interactive Dashboard & Visualization
Microsoft Excel	Initial File Inspection
CloudConvert	Independent XLSX to CSV Conversion
Anthropic Claude	Data Verification & Workflow Guidance
Canva	Report Design & PDF Creation
GitHub	Version Control & Public Project Repository
Notion	Process Documentation

About the Author

John Miguel Co Molina is a Remote Operations Support professional and junior data analyst based in Laguna, Philippines, with a developing focus on aviation and aerospace data analytics. This project represents the first entry in an ongoing portfolio series dedicated to surfacing actionable insights from publicly available aviation safety datasets.

With a professional background in remote operations support, John Miguel brings a systems-oriented perspective to data analysis – an understanding of how operational decisions are made in real time, and what kinds of data insights are most useful to the people making them. His portfolio work applies this operational lens to aviation data, combining SQL-based exploration, interactive dashboard development, and professional stakeholder communication to produce analysis that is both technically rigorous and practically relevant.

This project is the first in a planned series of aviation data analytics projects. The second project currently in development is the Global Aviation Footprint – a geospatial analysis of airport density and distribution patterns across all continents using the OurAirports dataset (57,000+ records), exploring how airport concentration varies by continent, elevation, and airport type.



This analysis was conducted independently. All data is sourced from publicly available government databases. No proprietary or confidential information was used in the preparation of this report.



github.com/cloudroseimage/faa-wildlife-strike-analysis



[linkedin.com/in/cloudroseimage](https://www.linkedin.com/in/cloudroseimage)



<https://linktr.ee/cloudroseimage>

"As long as birds and humans are flying, there are going to be collisions. But we have learned a lot about how to reduce those risks."

– *Carla Dove, Forensic Ornithologist,
Smithsonian Institution*